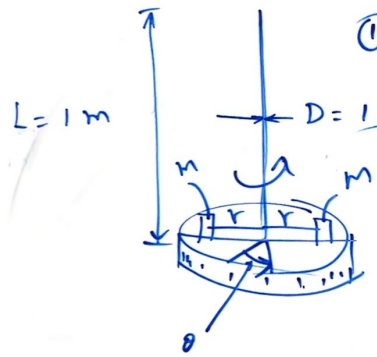
6-7-26

Waves Lec 8



① $C = \frac{\pi G D_{wire}^4}{32 L}$

$I = \frac{1}{2} M R_{disc}^2$

② $I = \frac{1}{8} M D_{disc}^2$

③ $\omega = \sqrt{\frac{C}{I}}$

$\tau = -C\theta = I \frac{d^2\theta}{dt^2}$

$\left(\frac{d^2\theta}{dt^2} = \ddot{\theta} = -\boxed{\frac{C}{I}}\theta\right)$ ω^2

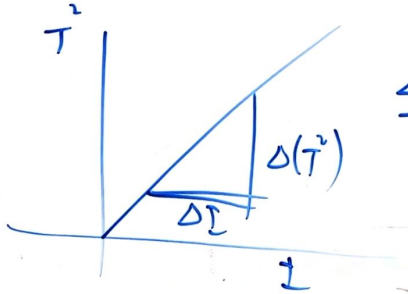
④ $\nu = \frac{\omega}{2\pi}$, $T = \frac{1}{\nu}$

$T^2 = 4\pi^2 \frac{I}{C}$

$T^2 \propto I$

$= k I$

#	T	T^2	$I \approx I_0 + 2mr^2$
1	T_1	T_1^2	I_1
2	T_2	T_2^2	I_2
3	⋮	⋮	⋮
n	T_n	T_n^2	I_n



$\frac{\Delta(T^2)}{\Delta(I)} = k$

$G_{steel} \approx 8 \times 10^{10}$ Pa

$D_{disc} \approx 10$ cm $= 0.1$ m

$M_{disc} \approx 1$ kg

$D = 1$ mm $= 10^{-3}$ m

$C = \frac{\pi \times 8 \times 10^{10} \times (10^{-3})^4}{32 \times 1}$

$\approx 8 \times 10^{-3}$

$I \approx \frac{1}{8} \times 10^{-2}$

$\omega = \sqrt{\frac{C}{I}} \approx \frac{8}{\sqrt{10}} \approx 2.5$

$\nu = \frac{\omega}{2\pi}$

$\approx$

$T =$

Ch. 1. P 1

①

A = 5cm

x(0) =

x = A cos

Find ω,

$G_{steel} \approx 8 \times 10^{10}$ Pa

$D_{disc} \approx 10\text{ cm} = 0.1$ m

$M_{disc} \approx 1$ kg

$D = 1\text{ mm} = 10^{-3}$ m

$$C = \frac{\pi \times 8 \times 10^{10} \times (10^{-3})^4}{32 \times 1}$$

$$\approx 8 \times 10^{-3}$$

$$I \approx \frac{1}{8} \times 10^{-2}$$

$$\omega = \sqrt{\frac{C}{I}} \approx \frac{8}{\sqrt{10}} \approx 2.5$$

$$\nu = \frac{\omega}{2\pi} \approx \frac{2.5}{2\pi}$$

$$\approx \frac{1}{2.5}$$

$T = 2.5$ s

Ch. 1. P 1-11.

① 

$A = 5$ cm, $\nu = 1$ Hz

$x(0) = 0$.

$x = A\cos(\omega t + \alpha)$

Find ω, α

(b) At $t = \frac{8}{3}$ sec,

Find x, dx/dt, d^2x/dt^2

② 1-12

③ $\nu = 1$, $x(0) = 1$, $v(0) = 1$

Given $x(t)$ in the form

$x(t) = A\cos(\omega t + \alpha)$,

Find ω, A, α, and s.t.

$x(t) \stackrel{?}{=} \sqrt{2}\sin(2\pi t + \pi/4)$

plot the fn